

Departement Elektriese, Elektroniese en
Rekenaar-Ingenieurswese

Department of Electrical, Electronic and
Computer Engineering

Klastoets 2
VRAESTEL

Kopiereg voorbehou

Class Test 2
QUESTION PAPER

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Module EBN111
Elektrisiteit en Elektronika
April 2018

Module EBN111
Electricity and Electronics
April 2018



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Eksamensinligting:
Exam information:

Maksimum punte: <i>Maximum marks:</i>	16	Volpunte: <i>Full marks:</i>	15
Duur van vraestel: <i>Duration of paper:</i>	30 min + 10 min to complete OCR form <i>30 min + 10 min om OKH vorm te voltooi</i>	Oopboek / toeboek: <i>Open / closed book:</i>	Toe <i>Closed</i>
Totale aantal bladsye (hierdie blad ingesluit): <i>Total number of pages (including this page):</i>		8 + OCR	

BELANGRIK- IMPORTANT

1. Die eksamenregulasies van die Universiteit van Pretoria geld.
The examination regulations of the University of Pretoria apply.
2. **Alle vrae moet op die op die Optiese Karakter Herkenning (OKH) vorm ingevul word.**
All questions must be answered on the Optical Character Recognition (OCR) form.
3. Neem noukeurig kennis van die instruksies op die OKH vorm.
Take careful consideration of the instructions on the OCR form.
4. Antwoorde moet eenhede insluit!
Answers must include units!
5. **Finale antwoorde moet as reële getalle geskryf word, en nie as breuke nie.**
Final answers must be written as real numbers, and not as fractions.

Dosent(e): <i>Lecturer(s):</i>	1. D le Roux
	2. D Ramotsoela
	3. X Ye

INSTRUCTIONS

Do step 0 in the first 5 minutes of the session. (0-5 min)

Do step 1 in the first 30 minutes of the test. (5-35 min)

Do step 2 within the following 10 minutes. (35-45 min)

STEP 0 (-5 min to 0 min): Complete the front page of the OCR form

STEP 1 (0 min to 30 min): Do your own calculations on the QUESTION PAPER. The question paper **will not** be taken in at the end of the test. Write down the answers for each question on the PRACTICE ANSWER SHEET. Be sure to include the units. The PRACTICE ANSWER SHEET **will not** be taken in at the end of the test.

STEP 2 (30 min to 40 min): Carefully copy the answers from the PRACTICE ANSWER SHEET to the OCR form.

- The Assessment ID is: **2018EBN111C2**
- Use a mechanical pencil with 0.5mm HB lead to complete the form.
- Each cell should only have one character.
- Characters may not touch or cross the cell boundaries.
- **Do not use commas!** Please make use of full stops.

Examples

01	-	1	.	4	5					m	A						
02	3	4	0							v							
03	1	5	.	2													
04	-	1								Ω							

INSTRUKSIES

Doen stap 0 in die eerste 5 minute van die sessie. (0-5 min)

Doen stap 1 in die eerste 30 minute van die toets. (5-35 min)

Doen stap 2 binne die daaropvolgende 10 minute. (35-45 min)

STAP 0 (-5 min to 0 min): Voltooi die voorblad van di OKH vorm

STAP 1 (0 min to 30 min): Doen jou eie berekeninge op die VRAESTEL. Die VRAESTEL **word nie** aan die einde van die toets ingeneem nie. Skryf die antwoorde vir elke vraag op die OEFEN ANTWOORDBLAD. Maak seker om eenhede in te sluit. Die OEFEN ANTWOORDBLAD **word nie** ingeneem nie.

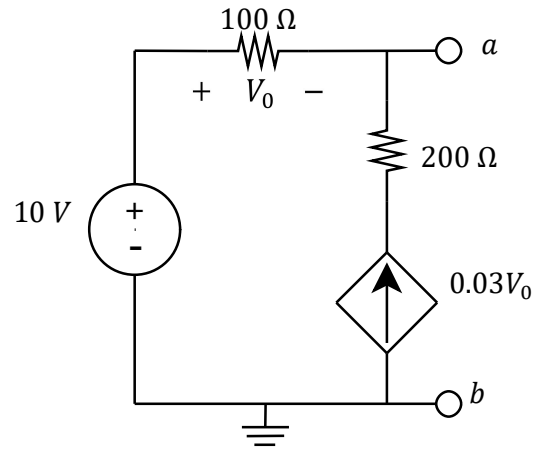
STAP 2 (35 min to 40 min): Dra die antwoorde vanaf die OEFEN ANTWOORDBLAD oor na die OKH vorm.

- Die "Assessment ID" is **2018EBN111C2**
- Gebruik 'n meganiese potlood met 0.5mm HB lood om antwoorde in te vul.
- Elke blokkie mag slegs een karakter bevat.
- Karakters mag nie aan die grense van die blokkies raak of oor dit steek nie.
- **Moenie kommas gebruik nie!** Maak eerder gebruik van punte.

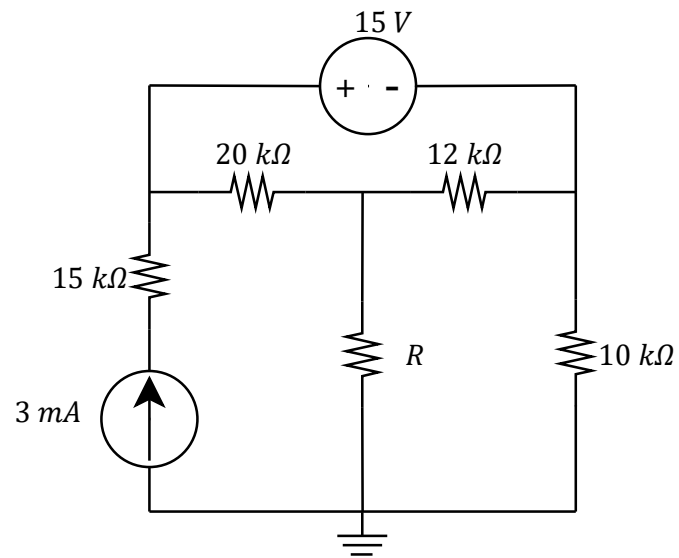
Voorbeelde

01	-	1	.	4	5					m	A						
02	3	4	0							v							
03	1	5	.	2													
04	-	1								Ω							

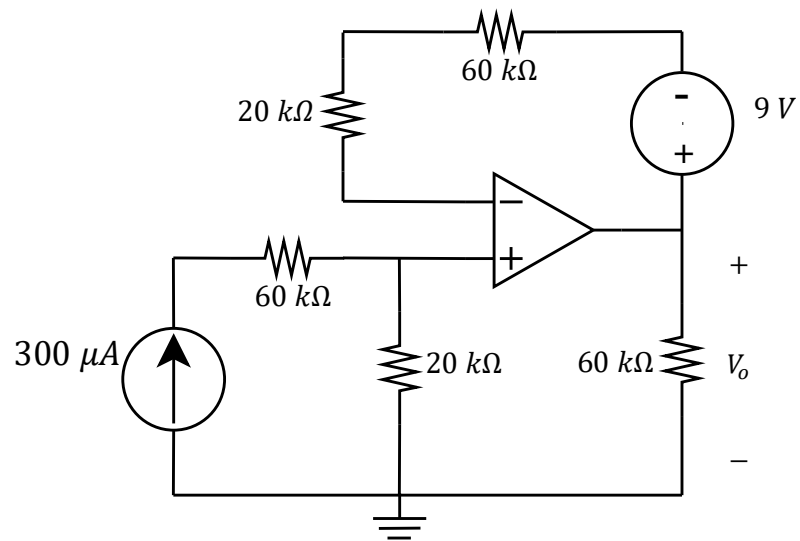
Questions 1-2 [4]		Vraag 1-2 [4]	
Determine the Norton equivalent current I_N and resistance R_N w.r.t. terminals a - b .		Bepaal die Norton ekwivalente stroom I_N en weerstand R_N met verwysing tot terminale a en b .	
01	I_N	01	I_N
02	R_N	02	R_N



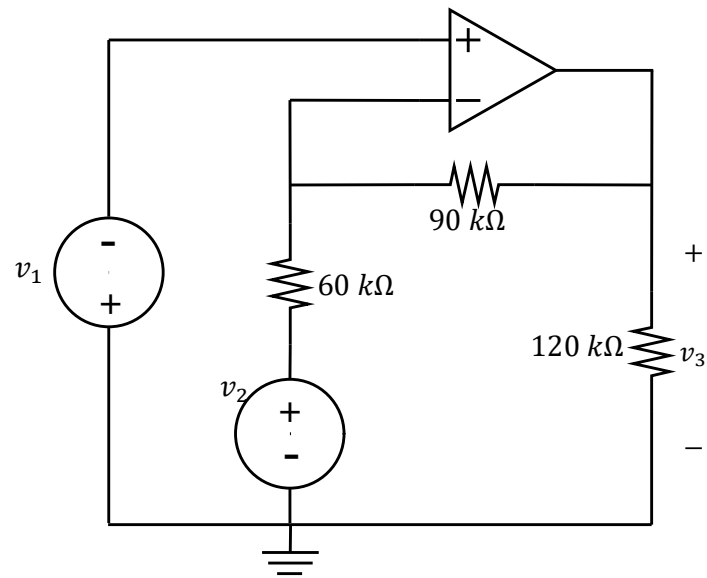
Questions 3 [2]		Vraag 3 [2]	
03	Find R such that maximum power is delivered to the $10\text{ k}\Omega$ resistor.	03	Bepaal R sodat maksimum drywing aan die $10\text{ k}\Omega$ resistor oorgedra word.



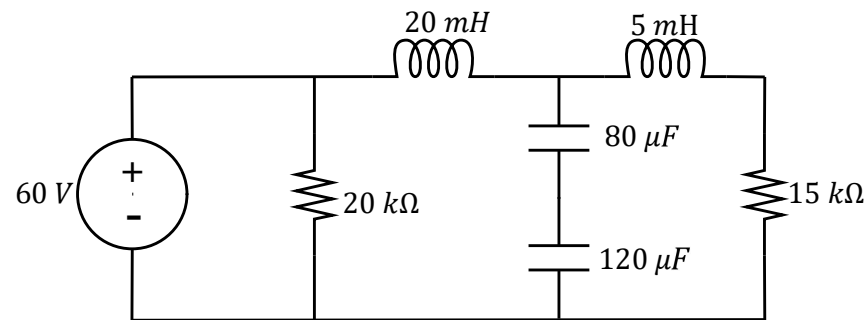
Question 4 [2]		Vraag 4 [2]	
04	Determine V_o .	04	Bepaal V_o .



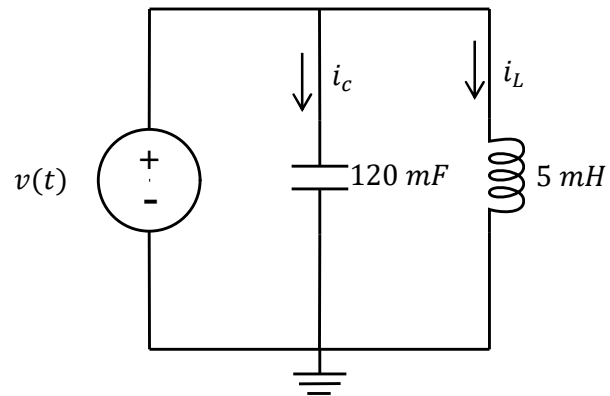
Questions 5-6 [4]		Vraag 5-6 [4]	
05	Determine v_2 if $v_3 = -12V$ and $v_1 = 0V$.	05	Bepaal v_2 indien $v_3 = -12V$ en $v_1 = 0V$.
06	Determine v_3 if $v_1 = 3V$ and $v_2 = 0V$.	06	Bepaal v_3 indien $v_1 = 3V$ en $v_2 = 0V$.



Questions 7 [2]		Vraag 7 [2]	
07	For the DC circuit below, determine the total energy w_C stored in the capacitors.	07	Indien GS-toestande aanvaar word, bepaal die totale energie w_C gestoor in die kapasitors.



Questions 8 [2]		Vraag 8 [2]	
Determine $i_C(2ms)$ if $v(t) = 20 \cos(50\pi t) mV$.		Bepaal $i_C(2ms)$ indien $v(t) = 20 \cos(50\pi t) mV$.	
08	$i_C(2ms)$	08	$i_C(2ms)$



PRACTICE PAGE / OEFENBLAD

Assessment ID

2	0	1	8	E	B	N	1	1	1	C	2		
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Answer

Unit

01	$I_N =$	
02	$R_N =$	
03	$R =$	
04	$V_o =$	
05	$v_2 =$	
06	$v_3 =$	
07	$w_C =$	
08	$i_C(2ms) =$	